Where, Not When: A Freeboard Analysis of Vehicle Flotation for Occupant Egress After Water Entry

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Abstract

Most occupants of cars that enter water survive the impact and drown afterwards, because the side windows—the only practical exit—go under water within about a minute. Automatic window opening is now rewarded by Euro NCAP and flotation bags have been patented, but to our knowledge no published work evaluates how long an opened window stays above water. We formulate post-entry flotation as freeboard regulation with a ground-truth metric, the Egress Freeboard Time (EFT). We prove that earlier buoyancy release is pointwise better in heave, so staging can only pay through information, and we derive a held-float gas threshold. We then test whether a closed-loop controller (FREE) that steers a gas reserve with hydrostatic waterline sensing beats one-shot flotation. In a 3-DOF truth simulator distinct from the controller's model, over 100 paired random vehicles per type, automatic windows alone succeeded in 8 % (ICE) and 0 % (EV) of cases. Flotation raised EV success to 89 % at 1.4 m^3. A one-shot, front-biased split tuned robustly on separate training draws matched FREE at low gas and beat it in the worst decile when gas was ample; feedback helped only when a rear bladder failed, and a failed front bladder defeated every design. Where the gas goes matters; when it is released does not, also at realistic fill times. A cost model shows that universal fitment exceeds accepted values of a statistical life at current incidence, so flotation suits high-exposure fleets near still water. All results are simulated.

Index Terms— Vehicle submersion, occupant egress, flotation, buoyancy control, hydrostatic sensing, functional safety, SOTIF, Euro NCAP.

I. Introduction

Occupants of cars that enter water usually survive the entry and then drown: in an autopsy series of 77 immersion deaths all but one were drownings and most victims had minor or no injuries [1]. In North America about 400 people die this way every year [2], and in the Netherlands submerged-vehicle crashes account for more than 50 deaths per year, 8.5 % of all road deaths [3]. Field tests define a *flotation phase* from water impact until water reaches the bottom of the side windows; it lasted 15–63 s in instrumented submersions, and escape through the side windows during this phase is the recommended strategy [2, 4]. Once water rises a short distance up the glass, power windows can no longer be lowered and doors cannot be opened against the external pressure [5].

Two engineering responses exist. Automatic window-opening systems detect submergence and lower the side windows [6–8]; Euro NCAP's 2025 Rescue & Extrication protocol now awards points when windows remain operable for at least 2 min after water entry or are lowered automatically within 2 min [9]. Separately, a family of patents adds buoyancy with inflatable bags or foam [10–15]. Both families act once, on a binary trigger; at most, a window-control patent chooses windows that are still above water at the moment of opening [16]. None manages or evaluates *how long the opened window stays above water*—the quantity that decides whether occupants can still climb out (Fig. 1).

This paper treats post-entry flotation as a *freeboard-regulation* problem and asks a design question: does a "smart" system that stages its gas and steers it by sensed waterline actually beat a well-designed one-shot system? Its contributions are:

1. A problem formulation whose objective is the Egress Freeboard Time (EFT): the time an automatically opened side window stays at least 10 cm above the free surface, evaluated on ground truth and linked to the Euro NCAP submergence criterion.
2. Three propositions: earlier buoyancy release is pointwise better in heave (so staging can only pay through information, not timing); a closed-form held-float gas threshold; and the centre-of-gravity range a gas reserve can correct.
3. FREE, a feedforward–feedback controller that steers a gas reserve with a multi-point hydrostatic estimate of the sill waterline, with a moving-water inhibit branch—the natural closed-loop design, implemented so that its value can be tested.
4. A reproducible simulation study with a 3-DOF truth plant distinct from the controller's model, open-loop baselines tuned on the truth plant both nominally and robustly on disjoint training draws, ablations, paired statistics, fault and robustness sweeps and a scenario catalogue. The study finds that automatic windows alone rarely leave enough time; that a robustly tuned, front-biased one-shot split matches closed-loop staging; and that feedback adds value only when a station fails.
5. A realism and impact assessment: a sensitivity study with the inflation times of real nitrogen floats, a mass, packaging and dynamics budget for the vehicle, a cost per life

saved against accepted values of a statistical life, and a map of the regions, industries and military uses where the system is most relevant.

Research questions: **RQ1** Can entry be classified quickly without deploying on fording or splashes? **RQ2** Does freeboard-steered staging increase EFT over the strongest one-shot strategy at equal gas? **RQ3** How much gas is required, and how does that scale for EVs? **RQ4** How robust are the designs to sensing errors, leaks and component failure? **RQ5** What does the system cost the vehicle, what safety does it buy, and where is it worth fitting?

Section II reviews prior work and states what is not new. Section III defines EFT, Section IV proves three properties of the problem, and Section V describes FREE. Sections VI and VII present the simulation study, Section VIII derives design rules, weighs the vehicle impact against the safety gained, identifies where the system pays, and discusses implementation and limitations, and Section IX concludes.

II. Related Work

A. Search protocol

Papers and patents were searched on 12–13 Oct 2026 (PubMed/PMC, ESV proceedings, Euro NCAP, preprint servers, publisher pages; Google Patents and USPTO full text) with seven query families: vehicle submersion and egress; vehicle flotation and inflatable buoyancy; submergence detection and window opening; staged inflation with attitude or water level; helicopter emergency flotation; vehicles in floodwater; Euro NCAP submergence. The last two years were covered in depth and seminal work included when cited. Engineering venues were covered through web-indexed queries restricted to IEEE Xplore, which returned no study of controlled vehicle flotation. Chinese utility models were sampled, not exhausted.

B. Human factors and epidemiology

Giesbrecht and co-workers established the three-phase model (floating, sinking, submerged) and the "seatbelts–windows–children–out" protocol [2, 4]; three adults and a child manikin exited one window in 53 s [2]. A 2024 systematic review lists vehicle safety enhancements among preventive measures [17]. Laminated side glass, adopted for ejection mitigation, defeats window breaking and makes powered or automatic opening essential [18]; Euro NCAP denies the tool route to laminated side glass [9]. Full-scale flood tests show a passenger car losing stability at about 0.40 m still-water depth [19], which bounds fording-detection thresholds.

C. Countermeasures and patents

Table 1 positions the closest prior art. Submergence detection [7, 8], orientation-gated automatic window opening [6, 20], flotation bags at four corners [10, 14, 15, 21], permanent foam [11], opening only windows that are not yet submerged after pressure-equalisation checks [16], a mid-window water-level trigger [14], flow-aware hazard sensing [12] and redundant immersion voting in helicopter floats [22] are all known; so is multi-stage inflator hardware [23]. Helicopter experience records asymmetric automatic float inflation as a cause of capsize [24], which is one reason symmetric all-at-once inflation is the conventional default.

Table 1: Positioning against the closest prior art (✓ = present; part = partially; implicit = not an explicit objective).

Capability	[10, 11]	[12]	[14, 15]	[6, 8]	[16]	[22]	FREE
Detect entry	–	✓	✓	✓	✓	✓	✓
Reject splash/flow	–	✓	part	✓	✓	✓	✓
Add buoyancy	✓	✓	✓	–	–	✓	✓
Open windows	–	–	–	✓	✓	–	✓
Window above water?	–	–	–	implicit	part	–	✓
Sill waterline estimate	–	–	–	–	–	–	✓
Reserve gas, staged	–	–	–	–	–	–	✓
Per-station feedback	–	–	–	–	–	–	✓
Objective = EFT	–	–	–	implicit	–	–	✓

What is not new here: detection, flotation bags or foam, four-corner placement, level-triggered deployment, automatic windows (including choosing windows that are still above water), flow-aware inhibition and redundant voting. **Claimed contribution:** to our knowledge no published work or patent found in this search uses the measured freeboard of the egress sills as the controlled variable for staged vehicle flotation, nor evaluates flotation by how long an opened window stays usable. We implement such a controller and test whether it is needed; the answer (largely no) and the theory explaining it are the main contribution.

III. Problem Formulation

Let $z_s(t)$ be the world height of sill $s \in \{\mathrm{FL}, \mathrm{FR}, \mathrm{RL}, \mathrm{RR}\}$ above the still-water surface, t_o the time automatic window opening is permitted (orientation gate $|\theta| < 30^\circ$, $|\varphi| < 20^\circ$ [20]), and $\varepsilon = 0.10$ m a usable margin. The objective is

$$\mathrm{EFT} = \inf\{t \geq t_o : \min_s z_s(t) < \varepsilon\} - t_o, \tag{1}$$

computed on the ground-truth plant only; $\inf \emptyset = \infty$, and EFT is capped at 300 s in the experiments (a held float). Fig. 1 illustrates the definition on two simulated runs. Success for n occupants is $\mathrm{EFT} \geq T_{\mathrm{req}}(n) = k\,[20 + 11(n-1)]$ s, anchored at the measured 53 s for four occupants [2]; the 20 s single-occupant value and linearity are assumptions. The benchmark uses $k = 1$; $k = 1.3$ models night or cold water (assumed) in the scenario catalogue.

The vehicle carries total gas V_g (volume at surface pressure) released at four stations at rates $u_s(t) \geq 0$ with $\sum_s \int u_s\,dt \leq V_g$. A *strategy* is a causal policy that maps the sensor history to $u_s(t)$; one-shot strategies release all gas at detection. Water inflow to compartment k through leak j follows an orifice law

$$\dot{V}_k = \sum_j C_d A_j \,\mathrm{sgn}(\Delta h_j) \sqrt{2g|\Delta h_j|}, \tag{2}$$

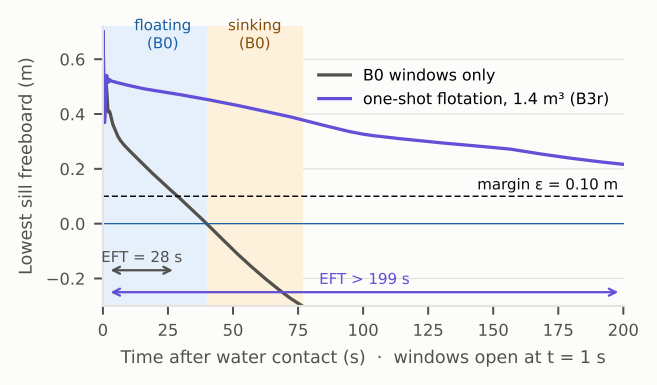


Figure 1: EFT on two simulated runs of the nominal ICE car. Without flotation (B0) the sill reaches the 10 cm margin 28 s after the windows open and the water line 40 s after contact (the calibrated floating phase, shaded blue). A one-shot flotation at 1.4 m^3 (B3r) keeps the sill above the margin for the whole 200 s run. Simulated.

where Δh_j is the difference between the external depth and the internal water depth at the leak point. Gas at depth d is compressed by $P_0/(P_0+\rho g d)$ (Boyle).

IV. Theory

Proposition 1 (Early release dominates in heave) *Consider heave only (no pitch or roll), one floodable compartment of footprint A_c with leak A at its floor that is not yet full, waterplane area $A_w > A_c$, and two buoyancy schedules with $B_1(t) \geq B_2(t)$ for all t. Then $V_1(t) \leq V_2(t)$ and the freeboard satisfies $z_{s,1}(t) \geq z_{s,2}(t)$ for all t.*

Proof: Draft is $d = (m/\rho + V - B)/A_w$ and the internal level is V/A_c, so $\dot{V} = f(V,B) = C_d A\sqrt{2g\max(0, d - V/A_c)}$. The function f is continuous, non-increasing in B, and—because $A_w > A_c$—non-increasing in V. Let $w = V_1 - V_2$ with $w(0) = 0$. Whenever $w(t) > 0$, $\dot{V}_1 = f(V_1,B_1) \leq f(V_1,B_2) \leq f(V_2,B_2) = \dot{V}_2$, so $\dot{w} \leq 0$; hence w can never become positive and $V_1 \leq V_2$ for all t (no Lipschitz condition is needed, which matters because $\sqrt{\cdot}$ is not Lipschitz at zero head). Freeboard $z_s = z_{\text{sill}} - d$ is decreasing in V and increasing in B, so $z_{s,1} \geq z_{s,2}$. ■

Corollary 1 *For a fixed total V_g, holding gas in reserve can never improve heave freeboard; any benefit of staging must come from* where *the reserve goes—information about pitch, roll and leak asymmetry that is unknown at detection.*

Proposition 2 (Held-float gas bound) *In the terminal state all floodable volume is flooded to the outside level. A level float with every sill at least ε above water requires*

$$V_g \;\geq\; V_g^{\text{nec}} = \frac{m}{\rho} - V_{\text{solid}}(d^\star), \tag{3}$$

with $d^\star = z_{\text{sill}} - \varepsilon$ and V_{solid} the non-floodable volume below $d^\star$.

Proof: Flooded compartments contribute no net buoyancy, so at equilibrium $m/\rho = V_{\text{solid}}(d) + V_{\text{gas,sub}}(d)$, where $d \leq d^\star$ is the draft. Because V_{solid} is non-decreasing in d and the submerged, compressed gas volume cannot exceed V_g, $m/\rho \leq V_{\text{solid}}(d^\star) + V_g$, which is (3). ■

Remark 1 (Sizing value) *At the margin draft $d^\star$ the gas is compressed by the factor $1 + \rho g d_b^\star/P_0$ (bladder-centre depth $d_b^\star$), so the gas that holds the float with the sills exactly at the margin is $V_g^{\min} = V_g^{\text{nec}}(1 + \rho g d_b^\star/P_0)$. Moment balance additionally fixes the front share.*

For the truth geometry ($V_{\text{solid}}(d^\star) = 0.31\ m^3$, compression factor 1.034) the necessary bound is 1.27 m^3 (ICE, one occupant) and 1.87 m^3 (EV); the sizing values are 1.31 and 1.94 m^3, rising by 0.078 m^3 per additional occupant. Section VII-C compares these thresholds with the simulated onset of held floats.

Proposition 3 (Correctable CG range) *If a fraction $1 - f_0$ of V_g is held in reserve and stations sit at $\pm x_b$, re-allocating the reserve between ends shifts the buoyancy moment by up to $\rho g(1-f_0)V_g x_b$, which corrects a longitudinal CG error up to*

$$|\delta x| \leq \rho(1-f_0)V_g x_b/m. \tag{4}$$

Proof: Moving reserve $R = (1-f_0)V_g$ from a symmetric placement to one end moves $R/2$ through $2x_b$, changing the moment by $\rho g R x_b$; equating with $mg\,\delta x$ gives the bound. ■ The bound neglects compression and partial submergence of the bladders. With $f_0 = 0.75$ and $x_b = 1.5$ m it is 0.33 m (ICE, 1.4 m^3) and 0.31 m (EV, 1.8 m^3), larger than the ± 0.1 m CG perturbation studied, so feedback is not needed to correct CG error in this range; its potential value lies in leak asymmetry and component failures, which Section VII-E tests.

V. Method: FREE

A. Entry classification and inhibit

Fig. 2 shows the system architecture and Fig. 3 the control flow. Deployment requires (D1) at least three of four underbody depth sensors above 5 cm for 0.5 s and (D2) loss of ground support on at least three wheels for 0.3 s; it is inhibited when a front–rear dynamic-pressure head above 2 cm or a drift speed above 0.3 m/s for 1 s indicates moving water. In moving water FREE reduces to automatic windows plus alert, because a buoyant car in a current is carried away (two of four recent fatal incidents we reviewed were of this type, Section VIII). The truth plant has no wheels and only still water, so the benchmark episodes use D1 alone (the vehicle floats by construction); D2 and the inhibit logic are evaluated separately in Section VII-H.

B. Waterline estimation

Four hydrostatic sensors at known body positions (x_i, y_i, z_0) report depths d_i. A plane $d(x,y) = a + bx + cy$ is fitted by least squares to submerged sensors (three suffice, so one failed sensor is tolerated); the sill freeboard estimate is $\hat{z}_s = -[a + bx_s + cy_s - \cos\theta\cos\varphi\,(z_{\text{sill}} - z_0)]$ using IMU attitude.

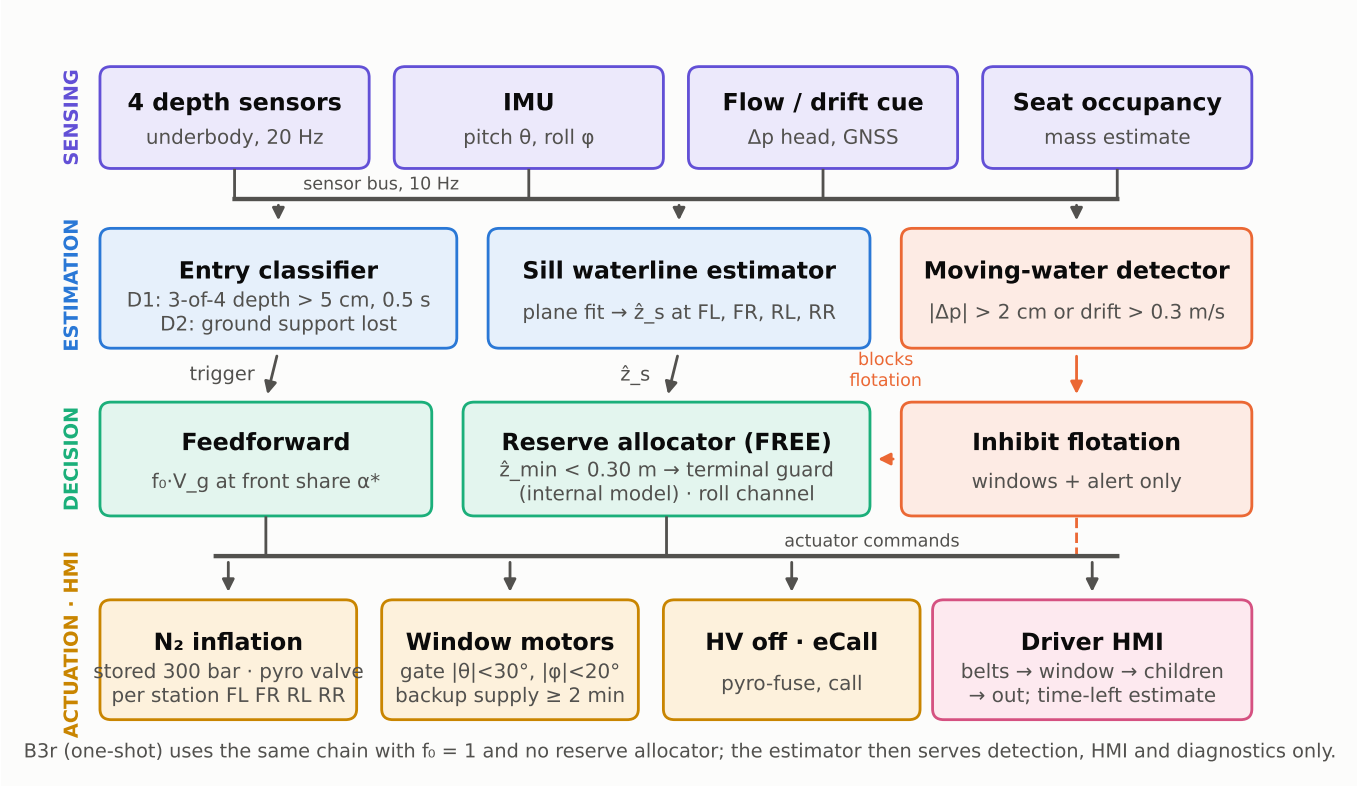


Figure 2: System architecture. Sensing feeds an entry classifier, a sill waterline estimator and a moving-water detector; the decision layer releases a feedforward share, steers the reserve (FREE only) or inhibits flotation; actuation covers stored-nitrogen inflation with one pyrotechnic valve per station, window motors on a backup supply, HV isolation, eCall and the driver HMI. The one-shot baseline B3r uses the same chain with $f_0 = 1$.

C. Feedforward and reserve

At detection FREE releases $f_0 V_g$ ($f_0 = 0.75$) at front share $\alpha^\star$—the nominal design-time split also used by baseline B3 (Section VI-B)—and holds the rest as $n_r = 4$ reserve charges of size $q = (1 - f_0)V_g/n_r$. Algorithm 1 releases a charge when $\min_s \hat{z}_s < 0.30\,\text{m}$ (refractory 1 s). A terminal guard uses a simple quasi-static box model with nominal parameters (deliberately different from the plant) to commit the whole reserve when it predicts a held float, or to choose the end; otherwise the end with the lowest sills is chosen, and within an end the lower side receives the larger share (roll channel). The guard costs at most $n_r + 3$ fixed-point solves per decision.

D. Design iteration (reported for transparency)

The pre-specified version (rev1, now ablation A0) released half of the gas symmetrically and steered the other half. A quick check against the truth-tuned one-shot baseline showed that holding half the gas back cost EFT, as Proposition 1 predicts, and exposed a bug in which a multi-charge commit loaded a single corner. The fix and the feedforward design ($f_0 = 0.75$, $\alpha^\star$ from the baseline's tuning) were fixed *before* the final benchmark, which uses fresh random seeds. After internal review, a robustly tuned open-loop baseline (B3r) was added; it is reported in full because it changes the conclusion.

Algorithm 1 FREE reserve allocation (every 0.1 s after detection)

if $n_r = 0$ **or** $t - t_{\text{last}} < 1$ s **or** $\min_s \hat{z}_s \geq 0.30$ **then**
  return
end if
$(k^\star, v^\star) \leftarrow \arg\max_{k \leq n_r} \textsc{Terminal}(V_F + kq, V_R + (n_r - k)q)$
if $v^\star \geq \varepsilon$ **then**
  fire $k^\star$ front, $n_r - k^\star$ rear (L/R split)
  return
end if
if guard predicts any float **then**
  end $\leftarrow$ end whose extra charge q gives the larger Terminal
else
  end $\leftarrow$ end whose sills have the lowest $\hat{z}_s$
end if
fire one charge at that end, on the lower side; $n_r \leftarrow n_r - 1$; $t_{\text{last}} \leftarrow t$

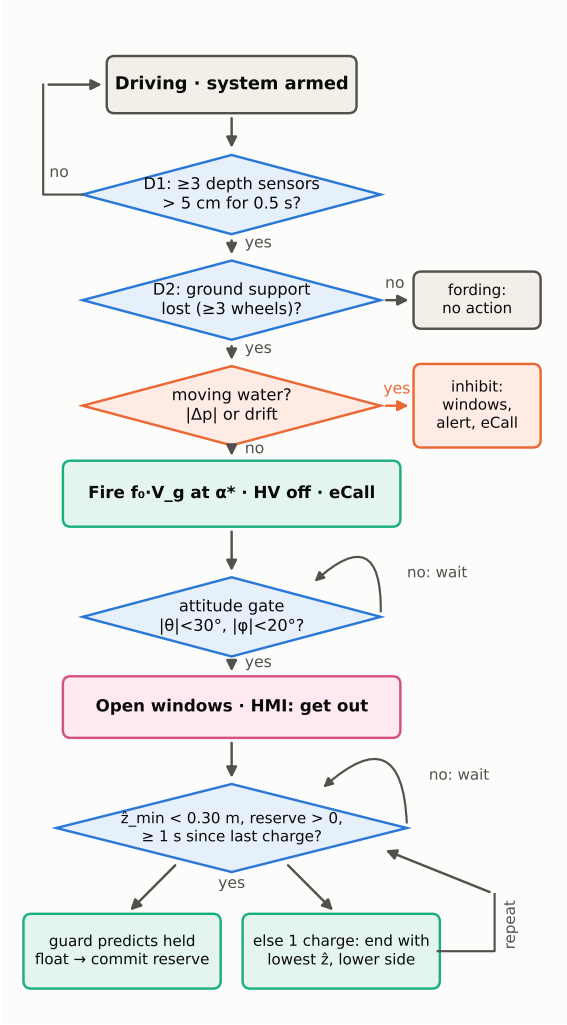


Figure 3: Control flow from the armed state to egress. Orange: moving-water inhibit; green: gas release; pink: window opening and HMI.

VI. Experimental Setup

A. Truth plant

The ground-truth plant (Fig. 4) is a 3-DOF (heave, pitch, roll) rigid body with a column-discretised tapered hull (approach and departure ramps, hood and boot lower than the cabin), three floodable compartments with tilted internal free surfaces, eight leaks (grille/underbody, floor and door seals left and right, boot, firewall and rear-seat pass-throughs) obeying (2), added heave mass equal to vehicle mass, damping ratio 0.3, and four outboard bladders (FL, FR, RL, RR) with first-order inflation lag τ_b and Boyle compression. Integration is semi-implicit Euler at 20 ms; control runs at 10 Hz. The controller never reads plant state; its terminal guard uses a different model (2-D quasi-static box hull, nominal parameters). Table 2 lists parameters; all are assumptions except the cabin leak area, calibrated once so that the nominal ICE car without a system reaches the sills after 40 s (centre of the measured 15–63 s [4]), giving 154 cm^2. With no further tuning the plant

Table 2: Main parameters ([A] assumed, [C] calibrated, [L] from literature).

Quantity	ICE	EV
Curb mass, CG ahead of centre [A]	1500 kg, 0.25 m	2100 kg, 0.05 m
Length × width [A]	4.6 × 1.80 m	4.7 × 1.85 m
Sill height above underbody [A]	0.80 m, sills at $x = +1.10/-0.90$ m	
Cabin leak area [C]	154 cm^2 (split floor/door seals)	
Grille/underbody leak, C_d [A]	0.20 m^2, 0.6	
Bladder stations, band [A]	$x = \pm 1.5$ m, $z = 0.10$–0.60 m	
Inflation lag τ_b [A]	0.15–0.50 s (random)	
Depth sensors [A]	4, $\sigma = 1$ cm, bias $\sigma = 1$ cm, 0.1 s latency	
IMU noise [A]	0.5°	
Window gate [L]	$\lvert\theta\rvert < 30°$, $\lvert\varphi\rvert < 20°$ [20]	
Egress margin ε [A]	0.10 m	
$T_{\text{req}}(n)$ [L+A]	$20 + 11(n-1)$ s, 53 s at $n = 4$ [2]	

predicts 21 s for the EV, inside the measured range; no measured EV submersion was found for comparison.

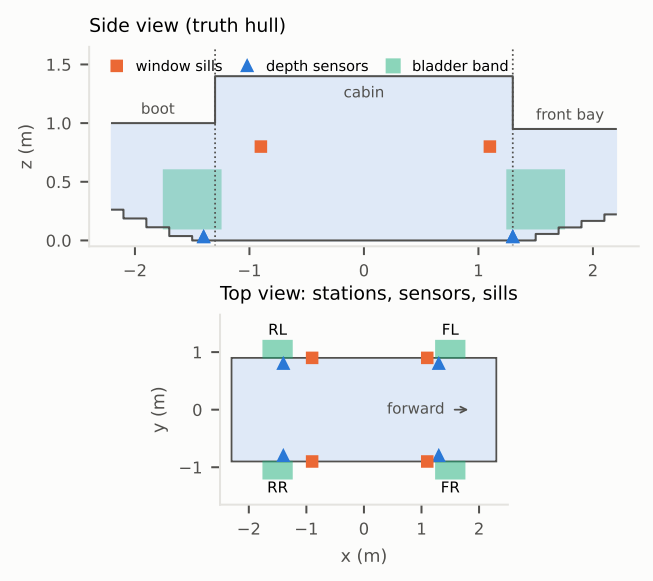


Figure 4: Truth-plant geometry (ICE): side view with compartments, sills, underbody depth sensors and bladder band; top view with the four stations. Simulated geometry, not a real vehicle.

B. Strategies

All strategies open the windows through the same gate. **B0**: windows only (the Euro NCAP-style automatic-window baseline). **B2**: one-shot symmetric inflation at detection (the behaviour of the flotation prior art). **B3**: one-shot at the front share $\alpha^\star$ that maximises EFT on the *truth* plant in the nominal scenario (grid 0.05). **B3r**: one-shot at the front share that maximises mean success over 30 training draws per vehicle (seeds 9000+, disjoint from the evaluation draws)—the strongest open-loop baseline; it chose α between 0.55 and 0.80, more front-biased than B3. **B4**: passive foam of the same volume at its own best front share (tuned like B3), present from $t = 0$, incompressible and without lag; a packaging-

unconstrained reference, not a practical alternative. **FREE**: as in Section V with $f_0 = 0.75$ and the B3 split (pre-specified). Ablations: **A0** the pre-specified rev1 (50 % symmetric, 50 % steered), **A1** no terminal guard, **A2** no roll channel, **A3** noise-free, zero-latency sensing; **FREEr** is FREE with the B3r split.

C. Monte Carlo design and statistics

Fig. 5 separates the information available at design time from the evaluation draws. For each vehicle, 100 random vehicles are drawn (seeds 5000+) and every strategy and gas level ($1.0, 1.4, 1.8, 2.2\,\mathrm{m}^3$) is run on the same draws (paired design, 8200 episodes): cabin leak $\times U(0.5, 2)$, left/right leak ratio $U(0.6, 1.4)$, front-bay and boot leaks $\times U(0.5, 2)$, mass ± 10 %, CG ± 0.10 m longitudinal and ± 0.05 m lateral, 1–5 occupants, entry pitch 0–30°, roll $\pm 5°$, entry speed 0.5–3 m/s, τ_b and sensor biases. No strategy is designed with access to the evaluation draws, so model mismatch is part of the test. EFT (1) is capped at 300 s (held float). We report medians and 10th percentiles (P10, the worst decile) with paired bootstrap 95 % CIs (2000 resamples), success rates, paired Wilcoxon signed-rank tests with Holm correction over all 80 FREE comparisons in the benchmark, matched-pairs rank-biserial effect sizes r_{rb}, and exact McNemar tests on success. Robustness and fault sweeps and the scenario catalogue use 40 fresh draws per cell (seeds 6000+, 7000+).

VII. Results

A. Simulator validation

Nine checks pass (Archimedes draft of a box hull within 0.1 %, static trim within 0.2 % of small-angle hydrostatics, zero roll for a symmetric case, exact water conservation for internal flow, compartment capacities respected, heave damping ratio 0.30 by logarithmic decrement, determinism, baseline invariance to FREE-only flags, and an inhibited FREE reproducing B0). The controller's internal model disagrees with the truth terminal state by up to 0.18 m near the held-float threshold (ICE, $1.4\,\mathrm{m}^3$), so the guard operates under real model mismatch (validation report in the repository).

B. Main benchmark (RQ2, RQ3)

Table 3 and Fig. 6 summarise the benchmark. **Automatic windows alone are not enough:** they succeed in 8 % (ICE) and 0 % (EV) of draws, with median EFT of 11 and 4 s, because the opened window is lost within seconds once the cabin floods. **Flotation changes this by an order of magnitude**, and the EV is the hard case: at $1.0\,\mathrm{m}^3$ symmetric inflation succeeds in 35 % of draws and the robust one-shot split in 78 %. Held floats appear only above the gas threshold of Proposition 2 (Section VII-C).

Where the gas goes matters; when it is released does not help. Against symmetric inflation (B2), FREE gains a median +29.2 s (EV, $1.0\,\mathrm{m}^3$) and +34.0 s (ICE, $1.0\,\mathrm{m}^3$) (Table 4). That gain comes from the front-biased split, which counters the nose-heavy flooding of the front bay, not from staging. A one-shot release at a robustly chosen split (B3r) matches FREE: the median difference FREE−B3r is +1.1 and -0.9 s for the EV at 1.0 and $1.4\,\mathrm{m}^3$, with success differences of +0.02 and +0.01. Where gas is ample FREE is worse in the tail: P10 -22.2 s [-31.6, -16.4] (ICE, $1.4\,\mathrm{m}^3$) and -17.0 s [-29.7, -9.1] (EV, $2.2\,\mathrm{m}^3$), because the 25 % reserve arrives later—exactly the cost Proposition 1 predicts. FREE's apparent advantage over the nominally tuned B3 (EV P10 +6.3 and +12.0 s at 1.0 and $1.4\,\mathrm{m}^3$) disappears against B3r: feedback was compensating for a sub-optimal nominal split, which robust tuning removes without any sensors. Fig. 7 summarises all paired comparisons with their confidence intervals.

Fig. 8 shows an example run per vehicle (full EFT distributions are in the repository). **Gas efficiency.** The gas needed for 90 % EV success is $1.96\,\mathrm{m}^3$ (B2), $1.64\,\mathrm{m}^3$ [1.36, 1.93] (B3), $1.45\,\mathrm{m}^3$ [1.22, 1.65] (B3r) and $1.40\,\mathrm{m}^3$ [1.15, 1.63] (FREE). These are interpolations on a four-level grid with bootstrap CIs; the B3r–FREE difference lies within them. For the ICE car all flotation strategies exceed 90 % at the smallest level studied, so efficiency is not resolved there. Larger occupant loads lower success, because both T_{req} and the gas threshold grow (Fig. 9).

C. Held-float onset versus theory

Fig. 10 compares the share of draws ending in a held float with the threshold band of Proposition 2 for one to five occupants. Held floats occur in 17–28 % of ICE draws at $1.4\,\mathrm{m}^3$ (inside the band 1.31–$1.62\,\mathrm{m}^3$) and in at least 92 % at $1.8\,\mathrm{m}^3$ (above it); for the EV they occur only at $2.2\,\mathrm{m}^3$, inside its band (1.94–$2.25\,\mathrm{m}^3$), in 34–47 % of draws. The closed-form threshold therefore predicts where held floats begin without simulation.

D. Ablations

Table 5 shows that, within FREE, the feedforward revision matters: the pre-specified pure-staging version (A0) is worse in every low- and mid-gas cell. The terminal guard (A1) changes nothing when no held float is reachable and helps the worst decile near the threshold. Neither the roll channel (A2) nor perfect sensing (A3) improves performance measurably, so we claim no benefit from roll control.

E. Robustness and faults (RQ4)

Sensor noise and latency hardly matter (sweep figure in the repository): with depth noise up to 6 cm and latency up to 1 s, FREE's EV success stays between 0.95 and 0.97. The feedforward fraction $f_0 \in \{0.5, 0.75, 0.9\}$ gives ICE median EFT 131, 152, 162 s against 163 s for one-shot release ($f_0 = 1$): the more gas is held back, the lower the median.

Fig. 11 compares fault tolerance against the strongest baseline. **A punctured rear bladder is the one condition where feedback clearly helps the ICE car**: success 0.97→1.00 and median ΔEFT +13.6 s for RL, and 0.95→1.00, +18.4 s for RR, because FREE re-routes reserve to the surviving rear station. For the EV the benefit is absent. With leaks two to three

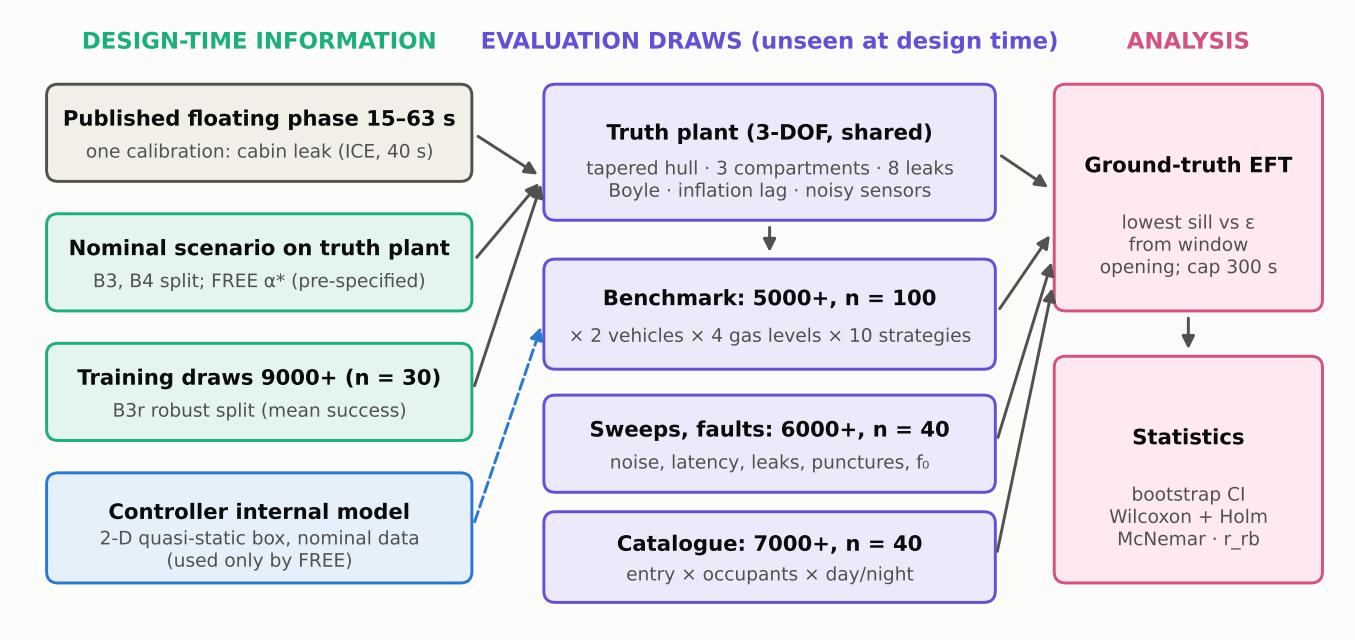


Figure 5: Study design. Everything on the left is available when a strategy is designed; the evaluation draws in the middle are never used for design or tuning. EFT is measured on the truth plant only, never on the controller's internal model.

Table 3: Benchmark (truth plant, 100 paired draws per vehicle): median/P10 EFT in s (300 = held float)/P(success). B0 (windows only): ICE 11/6/0.08, EV 4/2/0.00. All values simulated.

Veh.	Gas m^3	B2 one-shot symm.	B3 one-shot nominal split	B3r one-shot robust split	B4 foam (ref.)	FREE
ICE	1.0	77/54/0.93	108/76/0.99	111/82/0.99	112/82/1.00	110/81/0.99
ICE	1.4	129/83/0.99	170/107/0.99	165/119/0.99	191/119/1.00	154/97/0.99
ICE	1.8	300/300/0.99	300/300/0.99	300/300/0.99	300/300/1.00	300/300/0.99
ICE	2.2	300/300/0.99	300/300/0.99	300/300/0.99	300/300/1.00	300/300/0.99
EV	1.0	36/21/0.35	61/39/0.74	66/44/0.78	59/41/0.74	66/46/0.80
EV	1.4	56/36/0.69	76/47/0.84	84/58/0.89	83/54/0.89	84/59/0.90
EV	1.8	90/52/0.86	106/64/0.94	104/74/0.97	118/76/0.98	103/73/0.97
EV	2.2	195/89/0.96	195/89/0.96	184/109/0.97	300/112/1.00	158/92/0.96

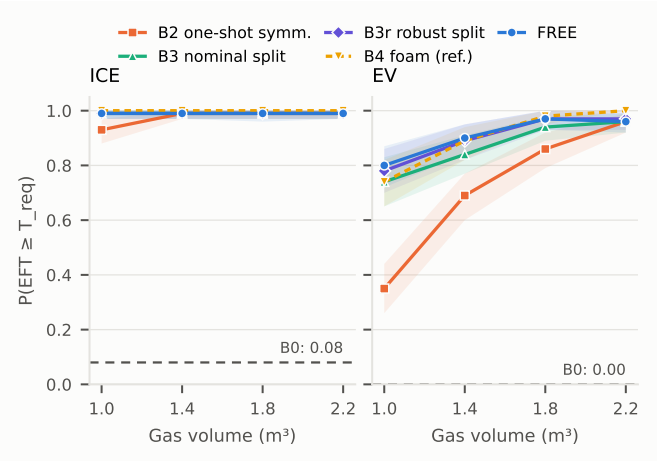


Figure 6: P(EFT $\geq T_{\text{req}}$) versus gas volume with bootstrap 95 % bands. Dashed yellow: passive foam reference B4; dashed grey: B0 windows only. Simulated.

Table 4: Paired differences FREE − baseline (s): median Δ vs B2; vs the strongest baseline B3r the median and P10 Δ [95 % CI], Δsuccess, r_{rb} and Holm-adjusted Wilcoxon p.

Veh.	Gas	vs B2	vs B3r median	vs B3r P10	ΔS	r_{rb}	p_{Holm}
ICE	1.0	+34.0	-0.7 [-0.9, 0.0]	-0.8 [-3.9, 2.3]	+0.00	-0.02	1
ICE	1.4	+15.0	-12.1 [-20.2, -7.5]	-22.2 [-31.6, -16.4]	+0.00	-0.60	0.00019
ICE	1.8	+0.0	+0.0 [0.0, 0.0]	+0.0 [-151.8, 0.0]	+0.00	-0.94	0.56
ICE	2.2	+0.0	+0.0 [0.0, 0.0]	+0.0 [0.0, 0.0]	+0.00	+0.00	1
EV	1.0	+29.2	+1.1 [0.0, 2.1]	+1.3 [-1.1, 4.2]	+0.02	+0.40	0.026
EV	1.4	+25.5	-0.9 [-1.3, 0.0]	+1.6 [-6.3, 3.2]	+0.01	-0.01	1
EV	1.8	+12.6	+1.1 [-3.2, 2.8]	-1.3 [-4.1, 3.2]	+0.00	+0.10	1
EV	2.2	-1.6	-6.7 [-14.4, 0.0]	-17.0 [-29.7, -9.1]	-0.01	-0.49	0.012

times nominal, B3r is at least as good as FREE. **A punctured front bladder defeats every design** (success $\leq$0.07): the nose-heavy flooding sinks a front sill first, and FREE, lacking a station-health input, even keeps sending reserve to the dead station. The front stations are a single point of failure and need redundancy.

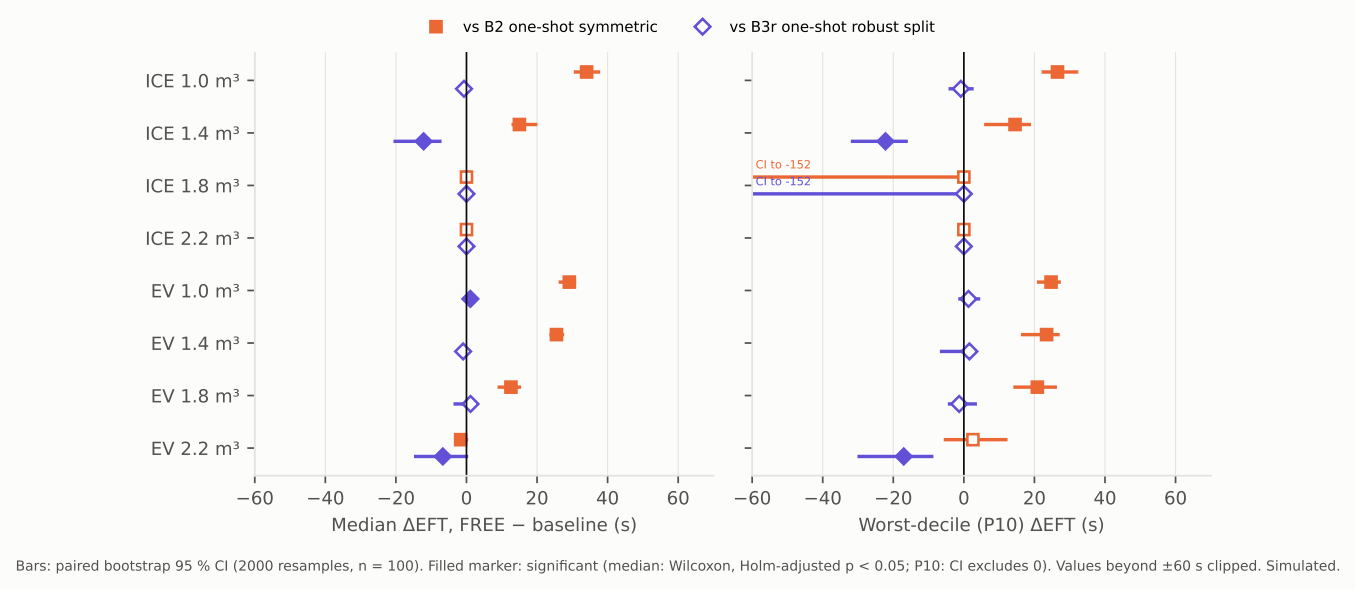


Figure 7: Paired differences FREE − baseline with bootstrap 95 % CIs: against symmetric inflation (B2, squares) FREE gains at low and mid gas; against the robust one-shot split (B3r, diamonds) the differences are near zero or negative. Filled markers are significant. Simulated.

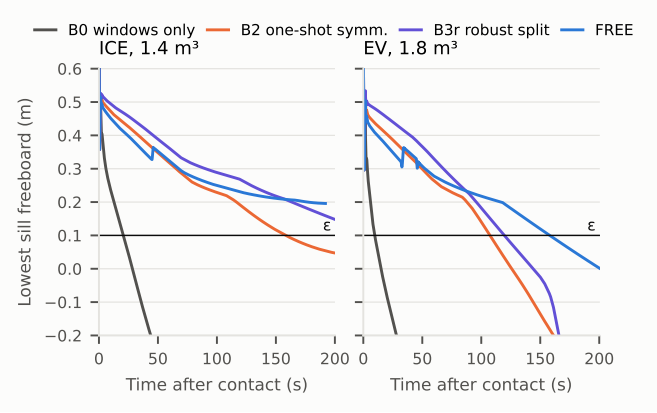


Figure 8: Lowest sill freeboard (truth) for one perturbed case per vehicle (leaks ×1.5, left-biased; case fixed before the analysis). EFT ends when a curve crosses 0.10 m; a line that stops early was classified as a held float. In this asymmetric case FREE outlasts B3r for the EV, which the 100-draw statistics do not confirm on average (Table 4). Single illustrative runs. Simulated.

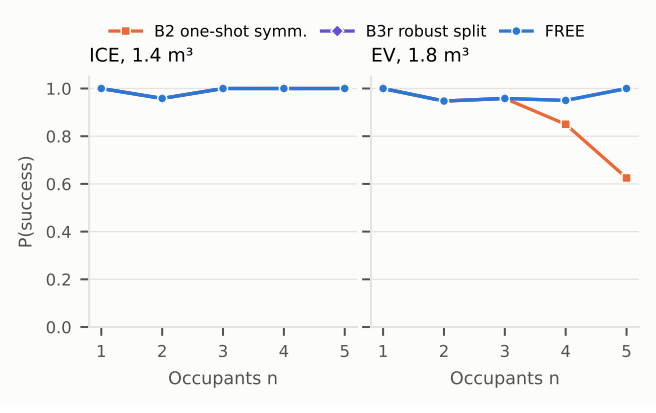


Figure 9: Success by number of occupants at mid gas (difficulty axis); B3r and, for the ICE car, B2 coincide with FREE where they are not visible. Simulated.

F. Scenario catalogue

Entry attitude (flat 0–10°, moderate 10–25°, steep 25–40°) is crossed with 1, 3 and 5 occupants and with day/warm versus night/cold water, the latter modelled as 30 % slower egress (assumed); Fig. 12 reports success at mid gas. Windows-only success is at most 0.00 in the EV cells and 0.12 in the ICE cells. Against B3r, FREE is higher in 3, equal in 13 and lower in 2 of 18 EV cells, and higher in 0, equal in 12 and lower in 6 of 18 ICE cells. Five occupants at night remain the hardest case (EV, flat entry: B3r 0.60, FREE 0.60). Entry attitude has little effect in this plant, because it does not model momentum-

Table 5: Ablations: median (P10) difference FREE − ablation in s. Positive = component helps.

Veh.	Gas	A0 rev1	A1 no guard	A2 no roll	A3 perfect
ICE	1.4	+17.4 (+7)	+0.0 (-8)	+0.0 (-0)	+0.1 (-0)
ICE	1.8	+0.0 (+143)	+0.0 (+128)	+0.0 (+0)	+0.0 (+0)
EV	1.0	+4.3 (+7)	+0.0 (+0)	+0.1 (+1)	+0.0 (+0)
EV	1.8	+5.7 (+3)	+0.0 (-0)	-0.2 (-0)	+0.1 (+1)
EV	2.2	+0.0 (-1)	+0.0 (-4)	+0.0 (-2)	+0.0 (+4)

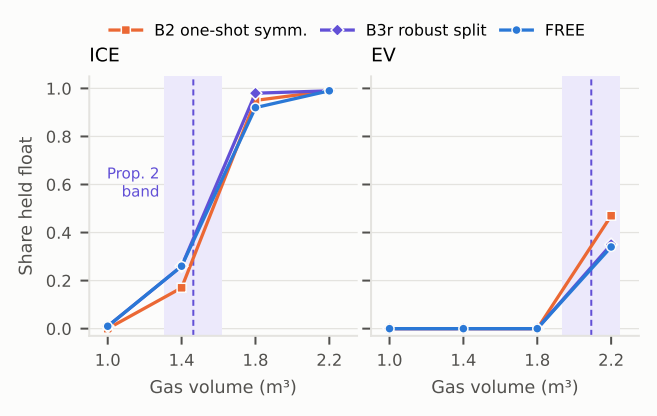


Figure 10: Share of draws with a held float versus gas volume; shaded: Proposition 2 sizing threshold for 1–5 occupants, dashed: three occupants. Simulated.

driven water ingress at steep entry and EFT starts at window opening—both limitations.

G. Realistic inflation time

The benchmark draws the bladder fill time constant τ_b from 0.15–0.5 s, which is faster than real cubic-metre floats. Helicopter floats are inflated with stored nitrogen in 2–3 s [25]; a CH-46 blow-down prototype took 13–14 s to full form against a 7 s design target, and a cool-gas generator reached 1 psig in 4 s [26]. For a first-order fill (95 % at $3\tau_b$) this is $\tau_b \approx 0.7$–4.5 s. We therefore repeated the mid-gas comparison on the robustness draws (seeds 6000+, $n = 40$ per vehicle) with $\tau_b \in \{0.3, 1, 2, 4, 6\}$ s (Fig. 13). Across this range the median EFT of B3r changes by at most 1.8 s and that of B2 by 2.2 s; FREE loses up to 13 s (ICE) at $\tau_b = 6$ s because its terminal guard predicts with the commanded rather than the realised volume. Success stays at or above 95 % for every strategy and τ_b. The reason is physical: gas is released at detection, while the car still floats on its own trapped air (40 s to the sills for the calibrated ICE car, about 21 s for the EV), so a fill of a few seconds is absorbed by the hull's own floating time. The comparative findings therefore hold for realistic stored-gas hardware; steep entries with fast momentum-driven ingress, which the plant does not model, would be more sensitive.

H. Entry classification (RQ1)

On synthetic event signals—a logic check, not evidence of field performance—the classifier deployed in 100 % of still-water entries (median latency 0.93 s, P90 1.25 s) and 0 times in 800 negative events (fording up to 0.5 m, splashes, car wash, stranded on wheels). It wrongly deployed in 14 of 200 moving-water entries, all at slow currents where the dynamic-pressure cue is below threshold.

I. Failure cases

(i) One ICE draw (83) fails for every strategy: the window gate opened during the entry plunge while the sill was momentarily within the margin; an AWOS-style 2 s delay or a freeboard condition on the gate would avoid this. (ii) Five-occupant cabins at night stay well below 90 % success at mid gas. (iii) Holding gas in reserve costs median and tail EFT whenever a one-shot design already holds a float. (iv) A failed front bladder defeats all designs and is not detected by FREE.

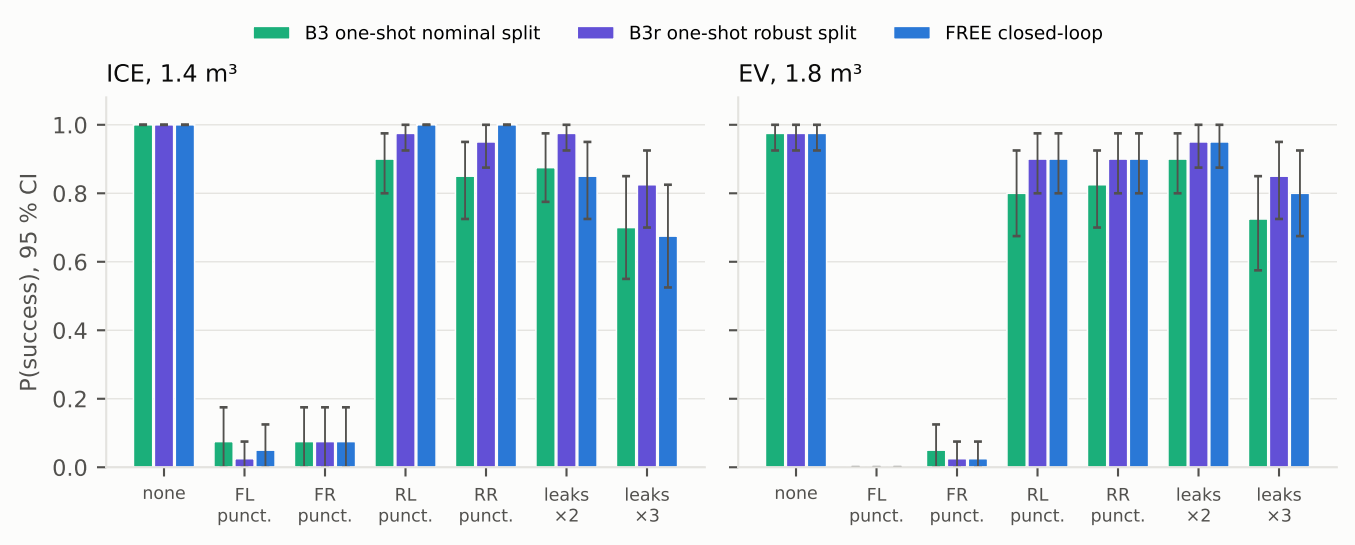


Figure 11: Fault tolerance at mid gas (ICE 1.4, EV 1.8 m^3; 40 draws): P(success) with bootstrap 95 % CIs for the nominal (B3) and robust (B3r) one-shot splits and FREE. Feedback helps only for a punctured rear bladder on the ICE car (FREE−B3r median ΔEFT +13.6 and +18.4 s, Holm-adjusted $p < 10^{-4}$); a punctured front bladder defeats every design. Simulated.

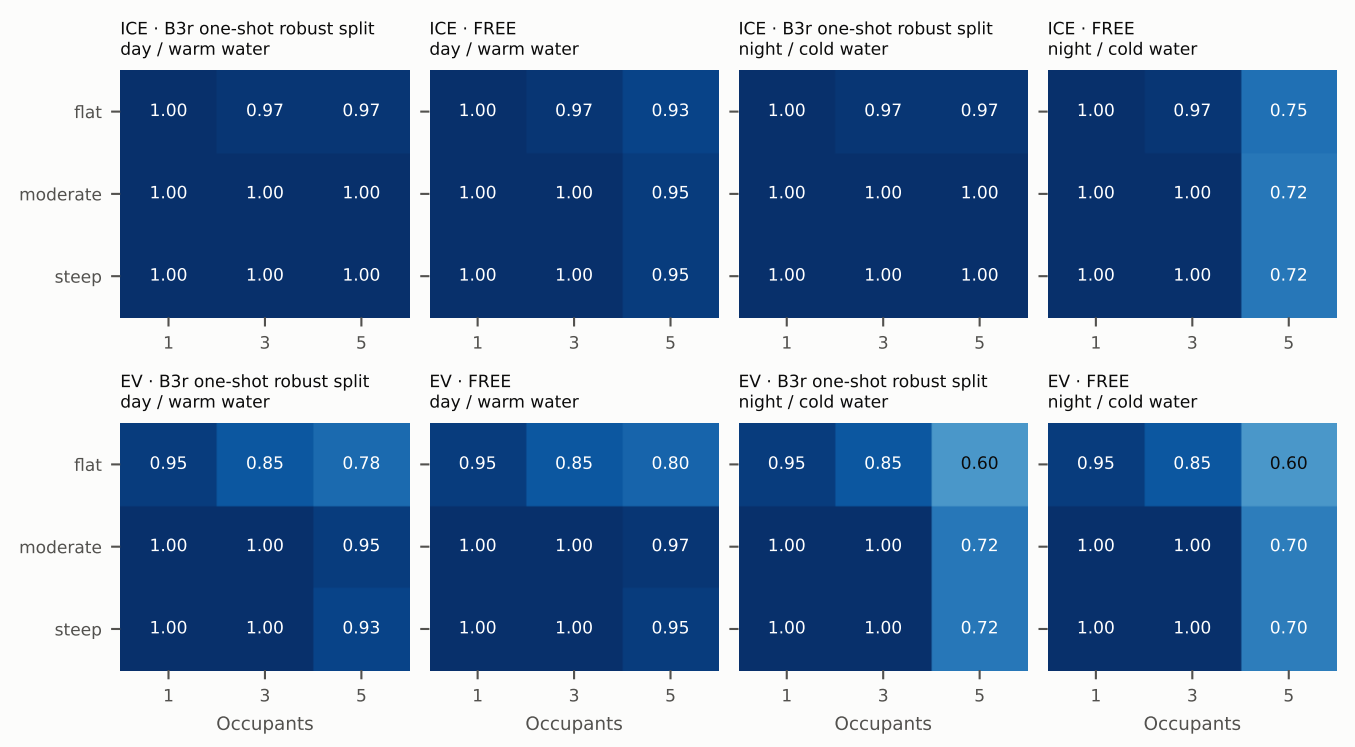


Figure 12: Scenario catalogue: P(success) for the robust one-shot split (B3r) and FREE by entry attitude and occupants, day/warm and night/cold (night factor assumed). 40 draws per cell. Simulated.

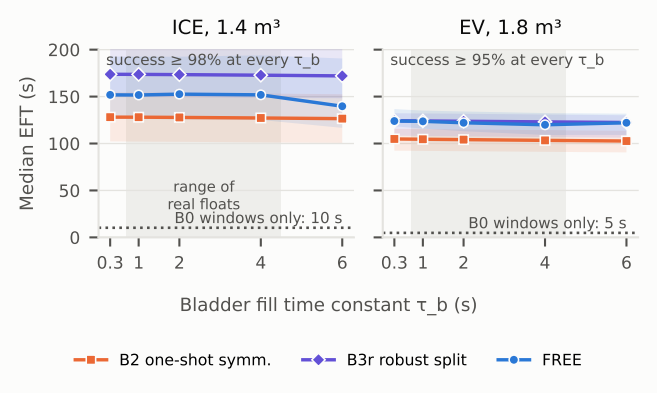


Figure 13: Realism check on inflation time (mid gas, $n = 40$ draws per vehicle, medians with bootstrap 95 % CI). Grey band: fill times of real nitrogen floats [25, 26]. Dotted: windows only (B0). The ranking B3r $\geq$ FREE $>$ B2 holds over the whole range.

VIII. Discussion

A. Design rules and the role of feedback

Design rules supported by this study. (1) Pair automatic windows with flotation; windows alone rarely leave enough time. (2) Size the gas with Proposition 2 for the heaviest expected cabin load plus margin. (3) Bias the gas towards the front and choose the split robustly over the expected spread of leaks, loads and entry attitudes, not for a nominal case. (4) Release it at once (Proposition 1). (5) Make the front stations redundant. Closed-loop staging is justified only as fault tolerance—re-routing gas when a station fails—and then only with a station-health input, which FREE lacked.

Why the "smart" design did not win. Proposition 1 removes any timing benefit, so feedback can only add information about where to put gas. Proposition 3 shows that a 25 % reserve can correct CG errors larger than those studied, but robust tuning already places the gas well for the whole distribution. What remains for feedback is the residual case-to-case variation, which turned out to be small next to the cost of late release. Robust open-loop tuning should therefore be the default comparison for adaptive safety actuators.

Role of sensing. Hydrostatic sensors remain essential for entry detection, the moving-water inhibit, a time-left estimate on the HMI and station-health monitoring; in the nominal case they are not needed to steer gas.

B. What the vehicle gives up and what it gains

Table 6 sets the cost to the vehicle against the safety gained for the mid sizing used throughout (ICE 1.4 m^3, EV 1.8 m^3). The nitrogen mass follows from the gas law; cylinder, fabric, housing and valve masses are engineering ranges (assumed, listed in `sim/impact.py`), anchored where possible on flotation hardware: four 1981 nitrogen cylinders of about 51 L in total weighed 37 kg [26], i.e. about 0.73 kg per litre, the upper end of our range. The system adds 15–32 kg, about 1–2 % of curb mass and less than one adult passenger. Its dynamic cost depends on packaging: with the cylinders near the centre of gravity and only the folded bladders at the corners, yaw inertia rises by about 1 %; with a cylinder at each corner it rises by up to 1.8 %, and the vessels would sit in the crumple zones. Central storage piped to the stations is therefore the realistic layout. For the EV, with an assumed 0.3–0.6 % energy use per 1 % of mass, range falls by under 1 %. Afloat, each corner mount carries 3.4–4.4 kN upwards, a load for which bumper-beam or subframe attachment points would have to be verified. The gain is large where the system applies: success rises from 8 % to 99 % (ICE) and from 0 % to 97 % (EV), and the median escape window from about 11 s and 4 s to more than 104 s.

Table 6: Vehicle impact versus safety gained (mid sizing). Basis: P physics, A assumed engineering range (to be verified), S source [25, 26], M this simulation (benchmark draws).

Quantity	ICE 1.4 m^3	EV 1.8 m^3	Basis
N_2 mass at bag pressure (kg)	1.8	2.3	P
Cylinder capacity, 300 bar (L)	7.5	9.7	P, A
Float fabric area (m^2)	12	15	A
Stowed volume (L)	22	27	A
System mass (kg)	15–28	17–32	P, A
share of curb mass (%)	1.0–1.9	0.8–1.5	A
Yaw inertia, central/corner cyl. (%)	+1.0/+1.8	+0.8/+1.4	A
Upward load per corner afloat (kN)	3.4	4.4	P
EV energy consumption (%)	–	+0.3–0.7	A
Fill time, real N_2 floats (s)	2–14	2–14	S
Gained: success, B0→B3r	8→99 %	0→97 %	M
Gained: median EFT (s)	11→165	4→104	M

C. Safety gained per unit cost

To ask whether the gain justifies the cost, we estimate lives saved per equipped vehicle as $r\,\phi_s\,\phi_u\,\phi_e\,\Delta P A$, where r is the submersion death rate per vehicle-year, ϕ_s the share of deaths in still or slow water (0.3–0.7, assumed), ϕ_u the share not incapacitated by the crash (0.45–0.70, anchored on 57 % of victims with minor or no injuries [1]), ϕ_e the share who leave when time suffices (0.5–0.9, assumed), ΔP the simulated success gain of B3r over windows alone, and $A = 11.9$ the discounted vehicle life (15 years, 3 %). Sampling these ranges, the system averts a median 18 % of submersion deaths among equipped vehicles (90 % interval 10–29 %). With a system cost of $300–1,200 (assumed), Fig. 14 gives the cost per life saved against exposure. In the Netherlands, the highest national exposure we found (more than 50 deaths per year [3] among 9.3 million cars [27], 5.4 per million vehicle-years), it is $63 M (90 % interval $26–137 M); for North America (about 400 deaths per year [2] over 298 million US vehicles [28], an upper bound of 1.3 per million) it is $254 M. Both exceed the US value of a statistical life, $14.2 M [34], and the European value of about €4.3 M [35]. The cost per life falls below these values only when exposure exceeds about 20 (US) or 60 (EU) deaths per million vehicle-years at $600, about four to eleven times the Dutch average. Two conclusions follow, and both survive the full uncertainty band. Fitting flotation to every car is not justified at today's incidence; it is justified for fleets with much higher exposure. Automatic windows, which cost little on cars that already have power windows, are justified everywhere—consistent with the Euro NCAP incentive [9].

D. Where the proposal is most relevant

The benefit model points to settings where water is still or slow, where vehicles are often near it, and where entries are frequent (Table 7). Countries with dense canal and polder networks, led by the Netherlands, have the highest national exposure we could document. Higher exposure is plausible in operator fleets that work next to water—port and dock vehicles, ferry ramps, dike and canal maintenance, dredging, flooded pits in quarries and mines, ice roads—and in police and rescue vehicles that are sent to the water's edge; we found no aggregate statistics for these fleets, and obtaining them is the first step of any business case. Floods are different: more than half of US flood deaths are vehicle-related [29], and at least 81 Australians died driving into floodwater in the 20 years to 2014 [30], but this is mostly moving water. There, added buoyancy lifts the car off the road sooner and helps the current

Table 7: Where post-entry flotation is most relevant. "Fit" combines exposure, water type and vehicle mass; quantified only where a source exists.

Setting	Evidence	Water	Fit of flotation
Canal and polder countries (e.g. Netherlands)	>50 deaths/yr, 8.5 % of road deaths [3]; 5.4 per million car-years [27]	still	highest national exposure; still below break-even for all cars
North America	≈400 deaths/yr [2]; ≤1.3 per million vehicle-years [28]	mixed	low per vehicle; windows first
Fleets working next to water (ports, ferry ramps, canal and dike maintenance, quarries, ice roads, police and rescue)	no aggregate data found; exposure plausibly far above national averages	still, ice	**best case**; first target for field data and pilots
Flash floods (US, Australia, Spain)	>50 % of US flood deaths vehicle-related [29]; ≥81 Australian deaths in 20 yr, 43 % of flood deaths [30]	moving	inhibit flotation; detection, windows and alert
Military tactical vehicles	123 training deaths 2010–2019, rollover in 63 % of fatal accidents [31]; 9 deaths at a flooded crossing [32]	moving	EFT metric, moving-water detection and egress HMI transfer; flotation needs tonne-scale gas
Buses on canal roads (e.g. India)	≥37 deaths in one event [33]; no national statistic found	still, slow	high consequence; exits above water rather than flotation

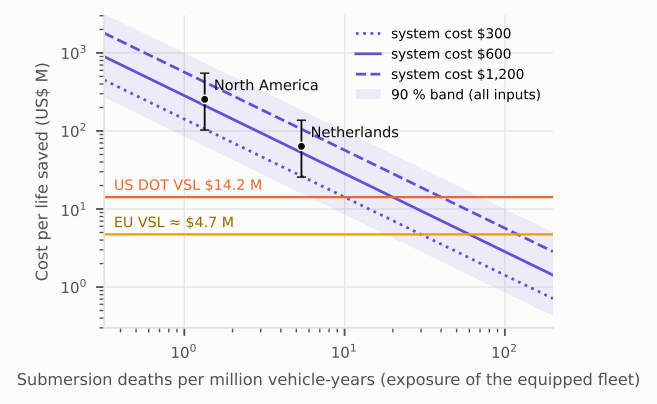


Figure 14: Cost per life saved against the death rate in the equipped fleet, for three assumed system costs (lines; band: 90 % interval over all model inputs). Points: national exposure for the Netherlands [3, 27] and North America [2, 28] at $300–1,200 (whiskers: 90 % interval). Horizontal lines: US [34] and EU [35] values of a statistical life (EU at an assumed 1.10 USD/EUR).

carry it, so the realistic contribution is detection, automatic windows and an alert, and the inhibit branch is essential. The same holds for military tactical vehicles: rollover was present in 63 % of fatal US Army and Marine Corps training accidents (123 deaths in 2010–2019) [31], and nine soldiers died when a truck overturned in a flooded low-water crossing in 2016 [32]. Such vehicles weigh several tonnes, so Proposition 2 scales the gas requirement accordingly, and the crossings involve current; what transfers is the EFT metric for hatch and window egress, the moving-water detector and the egress HMI. The sinking of an amphibious assault vehicle with nine deaths in 2020 [36] concerns the reserve buoyancy of a vehicle designed to swim, a different engineering problem. Buses that leave the road into canals cause the highest death tolls per event—at least 37 in one Indian case [33]—but would need an order of magnitude more gas than a car; for them, exits that stay above water matter more than flotation.

E. Realistic implementation

Gas source. Automotive airbag inflators are sized for cushions more than an order of magnitude smaller than 1.4–2.2 m^3, so the realistic source is stored nitrogen at about 300 bar released by a pyrotechnic valve, as in helicopter floats [25, 26]: 8–10 L of cylinder capacity in the mid sizing, with real fill times that do not change the findings (Section VII-G). A one-shot split (B3r) needs only a fixed orifice ratio and one valve per pair of stations; FREE needs an individually commanded valve per station, which adds cost and failure modes—a further argument for the one-shot design. Front redundancy can be provided by dual-chamber bladders.

Energy and windows. The window motors must work for at least 2 min after immersion [9], so they need a supply that survives water ingress into the 12 V network.

Unintended deployment. Deployment on the road is the dominant hazard: bladders out at speed would affect braking and steering. It needs a separate firing path with plausibility checks (D1 and D2, vehicle speed) [37]; ASIL values have not been assigned here. Triggering conditions (moving water, ice, inverted landing, laminated glass with failed windows, failed front station) form the SOTIF catalogue [38]; the slow-current misclassification is a known-unsafe scenario that needs a better flow cue.

Crash compatibility and service. Corner bladder packs sit in the crumple zones and must neither fire nor fragment in a collision; the vessels belong outside the crush zones. Cylinders need pressure monitoring and periodic requalification under the applicable pressure-vessel rules, and the system is replaced after deployment, as airbags are.

HMI. An opened window helps only if occupants use it. The display (seatbelts, window, children, out) mirrors the evidence-based protocol [2]; a freeboard estimate allows an honest time-left indication. Its effect on behaviour is untested.

Regulation. Euro NCAP rewards windows that stay operable for 2 min [9]. Our results suggest that operability alone is insufficient for heavy vehicles: the exit must also stay above

water long enough, which an EFT-based criterion could capture.

F. Threats to validity and limitations

Recent incidents. Of four fatal water-entry incidents reported in 2026 that we reviewed—a river entry in Sunderland, UK, with three deaths and one survivor [39]; flood sweep-aways in Spain [40] and Mississippi [41]; a harbour entry in California [42]—two involved moving water, where flotation must be inhibited. This anecdotal sample cannot estimate proportions, but it shows that the inhibit branch is not a corner case.

Threats to validity. All results are simulated, and only the floating time is calibrated; leak paths, cabin flooding, bladder packaging, gas dynamics and EV behaviour are assumptions. Absolute EFT values should not be quoted as predictions; the comparative findings and the theory are the contribution. The mass, cost and benefit factors of Sections VIII-B–VIII-C are engineering ranges, not measurements; the sign of the cost–benefit conclusion holds over the whole range, its magnitude does not. B3r was added after review; FREE was not re-tuned in response, and FREEr (FREE with the B3r split) performs similarly (repository table T7). EFT is counted from window opening, so time lost before the gate opens is not penalised; the gate is identical for all strategies. Ties at the 300 s cap reduce test power at high gas. The ICE gas grid starts above the efficiency threshold. Prior-art coverage of Chinese utility models is incomplete.

Limitations. Moving-water dynamics, shallow water in which the car grounds, waves and debris are not simulated; ice, inverted entry and occupant behaviour are out of scope; the classifier is evaluated only on synthetic signals. The tank, crane and test-track plan in the repository (1:10 Froude-scaled models with video ground truth, then scrapped vehicles) addresses these in order.

IX. Conclusion

Treating post-entry flotation as freeboard regulation gives a clear answer to a design question. Automatic windows are not enough on their own. Flotation sized by a simple buoyancy threshold, biased towards the front and released at once captures nearly all of the attainable benefit. A closed-loop, sensor-steered staging controller—the obvious "smart" design—did not beat a robustly tuned one-shot split: it was as good at low gas, worse in the tail with ample gas, and better only when a rear station failed. Where to put the gas matters; when to release it does not, and realistic fill times of a few seconds do not change this. The system adds about 1–2 % to curb mass; at today's national incidence, fitting it to every car would cost more per life saved than accepted values of a statistical life, so its realistic market is fleets that operate next to still water, while automatic windows are justified everywhere. Physical validation and exposure data for those fleets are the next steps.

Declarations

Data and code availability: simulator, controller, experiment scripts, raw per-episode results and a validation notebook are provided in the accompanying repository (to be archived with a DOI on release). **Conflict of interest:** the author is employed by a Tier-1 automotive supplier; the work was done independently, uses no employer data or tools, and has not been reviewed by the employer. Employer IP clearance is pending and must precede public release. **AI use:** an AI assistant (Claude, Anthropic) was used for literature search support, code drafting, simulation, statistics and drafting of text; the author is responsible for all content and verified the references listed. **Funding:** none. **Ethics:** no human or animal participants. **Images:** all figures are synthetic; no identifiable persons.

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